

Article

Comparing XGBoost and Double Machine Learning for Predicting the Nitrogen Requirement of Rice

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Highlights

What are the main findings?

- Double Machine Learning outperformed XGBoost in identifying heterogeneous nitrogen yield responses under real farm conditions.
- A hybrid decision support system combining DR-Learner and Causal Forest models was developed for site-specific nitrogen management.

What are the implications of the main findings?

- Causal machine learning enables yield-oriented nitrogen recommendations by adjusting for confounding in observational field data.
- Integrating plant-status information with causal treatment effects improves the reliability of nitrogen response estimation.

Abstract

Estimating how crop yield responds to site-specific nitrogen (N) fertilization is essential for maximizing yield potential under variable field conditions. However, classical Machine Learning (ML) approaches applied to observational farm data primarily focus on yield prediction and often fail to recover causal N response due to confounding arising from non-random fertilizer application. In this study, we develop and evaluate a Causal Machine Learning (CML) framework to estimate heterogeneous N treatment effects under real commercial rice-farming conditions in the Axios River Plain, Greece. The proposed approach combines Double Machine Learning (DML) with remote sensing, soil, climatic, and management data to adjust for confounding and identify causal relationships between N inputs, Leaf Nitrogen Concentration (LNC), and yield. A doubly robust (DR) learner is used to estimate yield sensitivity to N at key agronomic thresholds, while a Causal Forest model leverages LNC to assess crop physiological N status. Results demonstrate that the CML-based framework outperforms conventional XGBoost predictive models in identifying field plots that are responsive to additional N. By integrating causal effect estimation with plant-status information, the proposed decision support system identifies zones where yield gains can be achieved through targeted N increases while avoiding overfertilization in non-responsive areas.

Keywords: rice nitrogen fertilization; double machine learning; XGBoost; precision agriculture; yield prediction; causal AI



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1. Introduction

Nitrogen (N) is the most significant macronutrient for rice growing, as insufficient supply limits yield [1]. Rice plants require N in large quantities, more than any other nutrient, and N fertilization represents the largest share of input costs for the rice growers. [2]. N is essential for achieving high rice yields because it stimulates vegetative growth and tillering. Conversely, excessive N application increases production costs, rice lodging risk, higher disease pressure, and N lost to the environment [3]. However, growers are reluctant to fertilize with low N doses, since any yield losses would outweigh the savings from lower fertilizer costs [4].

Water management in rice fields strongly influences N losses to the environment. In flooded systems, N is supplied primarily in the ammonium (NH_4^+) form (e.g., urea or ammonium sulfate) [5]. In the Axios River Plain, rice growers drain their fields twice to apply herbicides, which forces them to maintain shallow water depths so that fields can be drained quickly for weed control (personal communication with local rice growers, Axios River Plain, Greece). The high summer temperatures in Greece further encourage growers to keep water levels low to limit algal growth. These practices promote more aerobic soil conditions. Limited water availability in the Axios plain during hot summers, which has become a more frequent complication in recent years due to increasing temperature extremes, can also lead to intermittent flooding and further increase soil aeration [6–8].

Under aerobic conditions, ammonium is rapidly oxidized to nitrite (NO_2^-) and nitrate (NO_3^-), which enhances N losses through gaseous emissions such as nitrous oxide (N_2O) and dinitrogen (N_2) [5]. Iatrou et al. [9] demonstrated an almost linear negative relationship between rice yield and radar backscatter, which is inversely related to exposed water surface in the field, highlighting the strong link between sustained flooding and yield performance. Consequently, Greek rice growers often apply high N rates to compensate for the greater N losses in their fields. Fertilization rates in the Axios River Plain typically range from 240 to 280 kg N ha⁻¹, whereas the recommended rates in the USA are 100 to 200 kg N ha⁻¹, with high-yielding fields receiving around 200 kg N ha⁻¹ [1].

Despite substantially higher N inputs, rice yields in the Axios River Plain are comparable to those in the USA, consistently exceeding 9 t ha⁻¹ and frequently reaching 12 t ha⁻¹, making it the most productive rice-growing region in Europe. However, N rates need optimization, as they can certainly be reduced by accounting for field-to-field soil variability and year-to-year climatic conditions.

Iatrou et al. [3] developed an XGBoost algorithm that predicted rice yield with a Mean Absolute Error (MAE) of 0.73 t ha⁻¹. The model's concept was to fit a yield response curve across several N doses and identify the dose that maximizes yield. This optimal N dose was then applied to the rice fields using Variable-Rate Applicators mounted on tractors equipped with monitors containing shapefiles that specified the application rate for each pixel of the map.

Iatrou et al. [9] developed a Variational Autoencoder (VAE) to reconstruct the input data and used these representations as inputs for a LightGBM regressor. The model achieved a prediction error of 0.6 t ha⁻¹ and therefore outperformed the XGBoost model. However, the LightGBM–VAE model was trained on 4884 field observations, whereas the XGBoost model was trained on only 274 observations because the experimental field units in that study were larger. Moreover, the LightGBM–VAE approach achieved an average yield increase of 4.32% compared with fields grown without N recommendations. Despite being trained on a larger dataset, the results showed that the VAE-enhanced model better captured the effect of N variability on rice yield than the model trained on the raw input data.

The widespread availability of Machine Learning (ML) methods and large datasets has made causal inference increasingly practical. New tools in Causal Machine Learning (CML) have accelerated this progress, and CML is now becoming important across many scientific fields [10–12]. In medicine, for example, causal methods are essential because it is often unethical or impossible to experimentally manipulate factors that affect human health [13,14]. Moreover, prediction accuracy frequently improves when models incorporate causal relations, since deeper structural understanding enhances model generalizability. In general, CML is a broad methodological framework that uses ML algorithms to estimate causal effects [14].

CML has also become important in economics and finance. Consider a simple example: customer spending may increase during certain periods of the year. If a bank offers a special credit promotion during such a period, one must ask whether the increase in spending is due to the promotion or simply due to seasonal effects such as Christmas. Classical ML models may exploit spurious correlations and overfit, leading to incorrect policy recommendations when estimating the effect of the credit offer [12]. The same challenge arises in agriculture, where there is substantial variability in soil properties across fields and strong year-to-year climatic variation. This raises a key question: how can we be confident that applying more N to a rice crop will increase yield? For decades, agricultural research has relied on Randomized Controlled Trials (RCTs) to isolate the effect of treatments from confounding factors [15]. However, RCTs are often expensive, labor-intensive, and limited in their generalizability due to substantial variability in soil properties across plots and strong year-to-year variation in weather conditions [16–18]. In addition, crop varieties change frequently, requiring new RCTs to be established to evaluate treatment responses for each new genotype, further increasing the cost of experimentation. With recent advances, the application of CML becomes increasingly feasible for estimating treatment effects, such as the impact of N fertilization on crop yield, using observational field data. These datasets are collected from farmers' fields where N rates and other management practices are not assigned randomly but instead reflect decisions made by farmers or agronomists using their domain knowledge and practical considerations. As such, observational data capture real-world management variability but do not satisfy the randomization conditions required by RCTs. Nevertheless, CML techniques are specifically designed to uncover nonlinear and heterogeneous treatment–outcome relationships while adjusting for confounding, and they can extract meaningful causal signals from these datasets [19]. The resulting treatment effect estimates can then be used to inform management recommendations and guide future N application strategies.

Recent years have seen the emergence of CML frameworks in agricultural research. For example, Giannarakis et al. [20] proposed a CML-based approach for assessing agricultural land suitability and its effects on ecosystem outcomes under varying management practices. In parallel, CML methods have been applied to evaluate the causal impact of agricultural policies on chemical fertilizer use, including studies of organic-substitute fertilizer policies [21] and analyses of factors influencing fertilizer use decisions [22]. These studies indicate that CML has begun to gain traction in the analysis of agricultural data. However, while recent studies have demonstrated the potential of CML for estimating N treatment effects under controlled or simulated on-farm experimentation settings (e.g., Kakimoto et al., [23]), to the best of our knowledge CML has not yet been applied to estimate the causal effect of N fertilizer application on crop yield using observational commercial field data. The objective of the present study is to address this gap by developing a CML-based decision support system that provides rice growers with data-driven N fertilizer recommendations.

2. Materials and Methods

2.1. Study Area

The study was conducted in the Axios River Plain in northern Greece, a low-lying coastal region of approximately 22,400 ha that represents the main rice-producing zone of the country. The area has a typical Mediterranean climate with temperate summers that are suitable for rice cultivation, including indica-type genotypes. Rice is the dominant crop, covering about 75% of the cultivated land, and is commonly rotated with maize, cotton, and alfalfa. The alluvial soils of the plain are predominantly silty clay, poorly drained, and classified as Typic Xerofluvents. However, salinity levels remain low and do not pose a degradation risk, according to recent findings by Litskas et al. [24]. Rice is typically sown in mid-May and harvested in October, depending on grain moisture levels, which ideally range between 19% and 21%. The Axios River Plain consistently achieves the highest rice yields in Greece, averaging close to 10 t ha^{-1} , compared with a national average of 8.89 t ha^{-1} . For this study, an area of 89.2 ha belonging to the rice farm of Mr. Kostas Kravvas, located near the town of Chalastra (40.6265° N , 22.7307° E , 6 m altitude), was selected. The fields were continuously cultivated with rice using precision agriculture practices during the study period (2017–2021). The average field size was 3.18 ha (Figure 1).

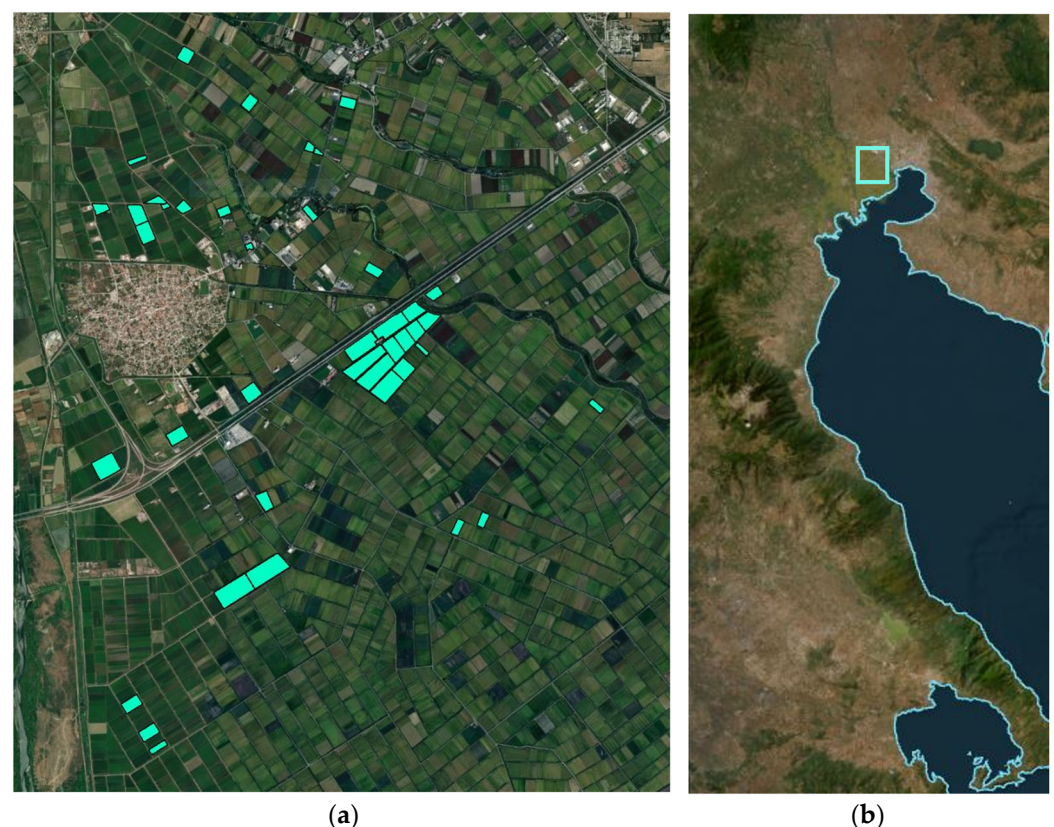


Figure 1. (a) High resolution view of the study area; polygons indicate experimental field parcels. (b) Regional overview map; the square denotes the location of the study area in the Axios River Plain, Greece.

2.2. Data Overview and Structure

The dataset used in this study integrates soil, climatic, remote sensing, management, and yield information collected from a commercial rice-farming system in the Axios River Plain, Greece, over five growing seasons (2017–2021). The spatial analysis was conducted on a regular $30 \text{ m} \times 30 \text{ m}$ (900 m^2) grid, resulting in 993 spatial units (0.09 ha per cell). After

temporal aggregation and data harmonization across years, the final dataset comprised 4884 plot-level observations. Soil data consisted of laboratory-measured physicochemical properties, including texture fractions, pH, electrical conductivity, organic matter, macronutrients, and micronutrients, derived from soil sampling conducted during the winter period. These point measurements were interpolated to continuous raster surfaces and extracted at grid-cell level. Additional soil-derived indices related to N availability and nutrient correction factors were computed using established agronomic mathematical equations [25,26]. Climatic data included temperature and precipitation variables obtained from satellite-based and reanalysis products, aggregated over agronomically relevant periods within each growing season. Remote sensing data were derived from Sentinel-1 and Sentinel-2 imagery, including synthetic aperture radar backscatter and optical indices used to estimate Leaf Nitrogen Concentration (LNC) [27]. All remote sensing variables were extracted at grid-cell resolution and aligned temporally with the crop growth cycle. Management data included N application rates from both broadcast and topdressing operations, implemented using variable-rate technology, and were recorded at high spatial resolution. Rice yield, used as the outcome variable, was obtained from calibrated combine-harvester yield monitors and subjected to standard cleaning and quality control procedures. For model development, variables were organized into outcomes (rice yield), treatments (N rate and LNC), confounders, and effect modifiers, following the CML framework. This structured dataset formed the basis for both causal and predictive modeling approaches described in subsequent sections. This formulation explicitly frames the problem as a multimodal CML task, integrating heterogeneous data sources to estimate treatment effects under real-world farming conditions.

2.3. Soil Properties

Cylindrical soil cores (30 cm depth) were collected from bare 212 soil surfaces during winter and analyzed at the Soil and Water Resources Institute, Hellenic Agricultural Organization (DIMITRA). Each sample was analyzed for 18 soil properties, including sand, clay, and silt percentages, bulk density, pH, electrical conductivity (EC), organic matter (OM), CaCO_3 , nitrate-N, phosphorus, potassium, magnesium, iron, zinc, manganese, copper, boron and calcium. The pH and electrical conductivity (EC) were determined using the saturated paste method [28]. Nitrate-nitrogen was extracted with a 2 M potassium chloride (KCl) solution and quantified using a Hitachi U-2900 UV-VIS spectrophotometer (Hitachi High-Tech Science Corporation, Tokyo, Japan) [29]. Organic matter was measured using the Walkley-Black procedure [30]. Calcium carbonate (CaCO_3) content was determined by titration [31], while soil texture was analyzed using the Bouyoucos hydrometer method [32]. Phosphorus was assessed following the Olsen extraction method [33]. Sodium, potassium, calcium, and magnesium were extracted using the ammonium acetate method and quantified via Inductively Coupled Plasma Spectroscopy (ICP) [34]. Micronutrients including Mn, Cu, Fe, and Zn were extracted with DTPA and measured using ICP [35]. Boron was extracted with the hot 0.02 M CaCl_2 method, which involves boiling the soil suspension for 5 min, and the extracted B was subsequently quantified using ICP [36]. The point-sample data were subsequently interpolated into raster surfaces for all measured soil properties using the inverse distance weighting (IDW) method, resulting in 18 continuous soil property maps.

2.4. Climatic Variables

Land Surface Temperature (LST) values were retrieved from the MODIS sensor using the MOD11A1 V6.0 product, which provides minimum, maximum, and mean temperatures at 5-day intervals [37]. Six-hourly temperature data for the study period were obtained

from the ECMWF ERA5-Land dataset ($0.1^\circ \times 0.1^\circ$ spatial resolution), from which daily sums, as well as minimum, maximum, and mean values, were computed and aggregated into 5-day intervals. Hourly precipitation data were retrieved from the Integrated Multi-Satellite Retrievals for GPM (IMERG) [38] at a $0.1^\circ \times 0.1^\circ$ spatial resolution. Accumulated daily precipitation and 5-day summary statistics (sum, minimum, maximum, and mean) were then calculated. All climatic variables were collected for the period of 15 May to 31 August for each growing season from 2017 to 2021.

2.5. Farming Practices

Fertilization events consisted of both broadcasting and topdressing applications. Throughout the study period, all applications were performed using variable-rate technology (VRT) with a Kverneland Geospread system equipped with a Tellus Pro terminal. This system allows for fertilizer spreading to be adjusted in 1 m sections with high precision. Prescription maps containing the fertilization recommendations were uploaded to the terminal as XML files, derived from GIS-generated shapefiles in which each polygon represented a management zone. Over the study years, these zones evolved from relatively simple configurations (2017–2019) to more complex ones (2020–2021). In 2017 and 2018, fertilization recommendations were based on expert knowledge and information from the literature [1,39].

2.6. Crop Spectral Properties

In this study, optical data was mainly used for the extraction of Leaf Nitrogen Concentration (LNC), an important indicator of rice crop photosynthetic capacity [27]. LNC was estimated from Sentinel-2 imagery obtained from the Copernicus Open Access Hub (copernicus.eu) in Bottom-of-Atmosphere (BOA) reflectance mode for multiple dates across the 5-year study period. The 2017 Top-of-Atmosphere (TOA) images were atmospherically corrected using the Sen2Cor algorithm within the SNAP/STEP software environment (Sen2Cor—STEP v.2.3.1 (esa.int)).

The LNC estimation equation was originally developed using hyperspectral data by Stroppiana et al. [40] and was later calibrated for Sentinel-2 by Karydas [41]:

$$\text{LNC} = 4.060 \times \text{NDRE} + 0.43,$$

where NDRE is the Normalized Difference Red Edge Index $(R_{790} - R_{720}) / (R_{790} + R_{720})$, with R_{720} corresponding to the 720 nm reflectance (Sentinel-2 Band 5) and R_{790} to the 790 nm reflectance (Sentinel-2 Band 7). Alongside the optical data, SAR backscatter in VH polarization was extracted for the same grid points from Sentinel-1 imagery acquired between 31 May and 24 June for each study year.

2.7. Yield Mapping

Rice yield was mapped throughout the study period using a 7.5 m Trimble yield monitor mounted on a Claas combine harvester, with automated guidance and field scanning (Trimble: 10368 Westmoor Drive, Westminster, CO 80021, USA; CLAAS: Harsewinkel, North Rhine-Westphalia, Germany). The raw yield data were first cleaned to remove unrealistic values (such as null, extremely low, or excessively high readings) and then calibrated using the field-level weighted yield measurements provided by the farmer, as mentioned by Karydas et al. [42]. Areas where yield was strongly affected by factors unrelated to fertilization, such as poor irrigation or rapid drainage, severe weed infestations, inadequate tillage, or application error, were excluded from the analysis. This exclusion was necessary because such unmeasured or uncontrolled factors could bias the relationship between N application and yield. Information on these sources of “noise” was obtained directly from

the farmer, whose detailed field knowledge is considered a reliable source of information on yield variability [43,44].

2.8. Experimental Design

A regular 30×30 m grid was generated across the study area, producing a point shapefile with 993 records in the GIS. The 0.09-ha resolution was deemed appropriate for capturing the spatial variability relevant to rice N requirements while avoiding excessive data volume and processing demands. The attribute table of the grid shapefile was progressively populated with soil properties, farming practices, and yield values by extracting the corresponding information from the original or processed geospatial layers. Climatic variables and crop spectral properties for each grid point were estimated using Google Earth Engine (GEE; <https://earthengine.google.com/>, accessed on 10 May 2022), a cloud-based platform capable of accessing and processing petabyte-scale archives of remotely sensed data [45]. The GEE catalog includes extensive satellite datasets, such as Landsat, Sentinel-1, Sentinel-2, Sentinel-3, MODIS, and ASTER, along with land cover, climatic, geophysical, and socio-economic layers. Data preprocessing in GEE was performed using JavaScript and geospatial Python (v.3.9) libraries [46,47]. Spectral index values were extracted for all clear-sky days (defined as <20% cloud cover) between 15 May and 31 August for each year from 2017 to 2021. When imagery was unavailable on a target date, or extraction was hindered by cloud cover, the most recent cloud-free image was used instead. Climatic and spectral variables were then joined to the grid shapefile. The resulting dataset was cleaned and harmonized through detailed visual inspection and geostatistical checks. The final dataset was used to train ML and CML models aimed at developing a rice yield prediction system from which accurate topdressing N recommendations for variable-rate technology (VRT) applications could be derived. Additional details on the data preparation can be found in Iatrou et al. [9].

2.9. Casual Machine Learning

The dataset used to develop the ML and CML models included additional variables that were not present in the work of Iatrou et al. [9]. The residual soil nitrogen (soil_resid_N) variable was calculated as the product of $(1 - \text{PHNL}/100)$, $(1 - \text{CCNL}/100)$, and $(1 - \text{DLNL}/100)$, following the approach of Koukoulakis and Papadopoulos [34]. Here, PHNL represents pH-induced N losses, CCNL denotes CaCO_3 -induced N losses, and DLNL corresponds to N losses due to denitrification and leaching. These components (PHNL, CCNL, and DLNL) are described in detail by Iatrou et al. [26].

The N inputs (N_inputs) variable was calculated by accounting for three soil N components: soil nitrate–N content, organic matter contribution, and soil texture effects, following the approach of Koukoulakis and Papadopoulos [25].

The first component, i.e., nitrate–nitrogen contribution ($\text{NO}_3\text{-N}_{\text{contr}}$), calculated as:

$$\text{NO}_3\text{-N}_{\text{contr}} = \frac{(\text{NO}_3\text{-N}) \times (\text{FEB} \times 1000 \times 300)}{10^6}$$

where $\text{NO}_3\text{-N}$ is the measured nitrate–N concentration and FEB is the soil bulk density factor expressed in g/cm^3 . This term estimates the mineral N available in the soil volume corresponding to the top 30 cm.

The second component, i.e., nitrogen mineralization contribution (NM_{contr}), represents the N contribution derived from organic matter mineralization:

$$\text{NM}_{\text{contr}} = \frac{(\text{FEB} \times 1000 \times 300) \times \text{OM}}{100}$$

where OM is the percentage of soil organic matter. This value represents the amount of organic material potentially contributing to N release.

The third term, i.e., soil texture contribution (Soil_Text_contr), accounts for the influence of soil texture on mineralization potential. Soil_Text_contr was assigned according to the soil texture class as follows:

Sandy soils: Soil_Text_contr = 0.013

Loamy soils: Soil_Text_contr = 0.015

Clayey soils: Soil_Text_contr = 0.018

The fourth component, i.e., organic matter contribution (OM_contr), represents the estimated N released from organic matter based on both BN and the texture-dependent CN value:

$$OM_contr = \frac{(NM_contr \times Soil_Text_contr)}{1.75 \times 12}$$

Finally, the total N inputs (N_{inputs}) available to the crop were calculated as the sum of mineral nitrate–N and organic N contributions:

$$N_{inputs} = NO3_N_contr + OM_contr$$

This integrated calculation reflects both the directly available nitrate fraction and the texture-modulated mineralizable N fraction, providing a quantitative estimate of soil N supply prior to fertilization.

The Phosphorus Fraction Percentage (PFP) variable, representing the percentage of applied phosphorus that was adsorbed by the soil, was calculated using the PyPI phosadsorption library at <https://pypi.org/project/phosadsorption/> (accessed on 17 October 2025) [48].

Finally, the dataset used to develop the ML and CML consisted of 4884 records. The data were randomly split into training and test sets, comprising 80% (3908 rows) and 20% (976 rows), respectively. To reduce dimensionality and eliminate non-informative predictors, a Random Forest algorithm was applied to calculate feature importance scores [49]. Variables with low importance were removed. The remaining subset of variables was then examined for collinearity using Spearman's rank correlation. Each time a collinear feature was considered for removal, the Mean Absolute Error (MAE) on the test set was evaluated. If removing a feature did not substantially increase the prediction error, the feature with lower importance according to the Random Forest scores was discarded [50].

2.10. Model Training

The final feature space comprised the following: (i) soil parameters measured by the soil laboratory; (ii) variables computed from soil mathematical equations; (iii) the remote sensing spectral index LNC; (iv) hindcast May precipitation data retrieved via Google Earth Engine (GEE; accessed 10 May 2022); and (v) climatic variables derived from seasonal forecast anomalies provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). Variables were organized into covariates (X), confounders (W), treatments (T), and the outcome (Y). Covariates, also referred to as effect heterogeneity features or effect modifiers, represent the variables that describe how the treatment effect varies across field plots. Confounders or control variables are those that influence both the treatment and the outcome. The outcome variable in all models was rice yield. The same feature space was used to fit both CML models and classical predictive models.

Two DML models were estimated: one using N as the treatment, hereafter referred to as N_DML, and one using the spectral index LNC as the treatment, referred to as LNC_DML. For the LNC_DML model, the treatment variable (LNC) was discretized into 10 classes using quantile-based binning. This procedure was adopted because LNC exhibits high variance when used as a continuous predictor of yield, a common characteristic of spectral indices and environmental variables. By transforming LNC into quantile-based classes, the influence of extreme values on conditional effect estimation was reduced, while ensuring that each class contained approximately the same number of observations [51]. Beyond variance stabilization, the primary motivation for binning was interpretability. This discretization enabled the construction of a clear sensitivity graph, allowing for the evaluation of how a one-class increase in LNC affected yield within the LNC_DML model and facilitating direct comparison with the corresponding predictive model.

Double Machine Learning (DML), used in the current study, is a specific subclass of CML methods designed to remove confounding bias and produce robust estimates of treatment effects [11,52]. Causal Forests were used for the LNC_DML model, while a doubly robust learner (DR-Learner) was used for the N_DML model. Causal Forests are well suited for estimating heterogeneous effects with continuous treatments, whereas DR-Learners are typically preferred when the treatment is binary or defined through thresholds [53,54]. In our study, LNC is available as a continuous treatment variable; however, for variance stabilization and interpretability, we transformed LNC into 10 quantile-based ordered classes. Accordingly, the LNC_DML model estimates the marginal yield response to a one-class increase in LNC status using a Causal Forest-based DML estimator. All LNC-related effects are therefore interpreted on the ordinal scale (per one-class increase), rather than per unit change in continuous LNC. In contrast, N was analyzed through two agronomic thresholds (240 and 260 kg N ha⁻¹), creating binary treatment indicators for each threshold. Therefore, a DR-Learner was used for the N_DML model.

The detailed procedures of each method (Causal Forest and DR-Learner) are outlined below:

2.10.1. Causal Forest

Conceptually, the Causal Forest DML procedure can be understood in four steps: First, a propensity model is fitted to predict the treatment (T) using confounders (W). Second, an outcome model is fitted to predict the outcome (Y) using the same confounders (W) and the covariates (X), which are potential effect modifiers. Third, residualized (orthogonalized) variables are computed:

$$\text{Residual treatment : } T^* = T - \hat{T}$$

$$\text{Residual outcome: } Y^* = Y - \hat{Y}$$

Finally, a simple regression of the residual outcome on the residual treatment is estimated:

$$Y^* = a + \theta T^* + \varepsilon$$

Because both the treatment and outcome have been purified from the influence of confounders, the coefficient θ provides a debiased estimate of the causal effect, orthogonal to the nuisance models. This orthogonalization step makes the estimator robust to high-dimensional controls and reduces sensitivity to overfitting relative to classical ML prediction models [11,55].

For estimating heterogeneous treatment effects for the LNC_DML model, a Causal Forest DML estimator was implemented using the EconML library developed by Microsoft Research [56]. The model was initialized with 4000 trees, min_samples_leaf = 5,

and $\text{max_depth} = 50$. The internal hyperparameters of the forest were tuned using the built-in tune routine of the CausalForest implementation in the EconML library. Nuisance models for the outcome and treatment were selected via a model-stacking approach using GridSearchCVList, which evaluated Lasso, Random Forest, and Gradient Boosting regressors over predefined hyperparameter grids using 5-fold cross-validation and negative mean squared error as the scoring metric. The optimal nuisance models for both $\hat{\mu}(X, W)(\text{model_y})$ and $\hat{e}(X, W)(\text{model_t})$ were Random Forest Regressors with $\text{min_samples_leaf} = 10$. These optimized models were then used to fit the final Causal Forest DML estimator on the training dataset.

2.10.2. DR-Learner

The doubly robust (DR) learner is a Causal Machine Learning method designed to estimate heterogeneous treatment effects by combining two complementary sources of information: (i) outcome regression and (ii) propensity-score correction. It is called doubly robust because it yields consistent estimates of the treatment effect if either the outcome model or the propensity model is correctly specified, which offers strong protection against model misspecification in observational settings [54,57].

The DR-Learner, used to fit the N_DML model, proceeds in several steps. First, a propensity model is fitted to estimate how likely each plot is to receive the treatment based on its observed characteristics. This step is essential because, in real-world agricultural data, treated and untreated plots often differ systematically. Without correcting this, one might incorrectly conclude that plots treated with high N levels should indeed always be treated more, even when this pattern merely reflects confounding factors such as farmers applying higher N rates to inherently more productive fields.

Second, two separate outcome models are trained: one using only treated observations to predict yield under treatment, and another using only untreated observations to predict yield under no treatment. These models approximate the two potential outcomes: what the yield would have been for each plot under treatment and under control.

Third, the DR-Learner combines the information from the propensity model and the two outcome models into a single doubly robust pseudo-outcome, following the DR-Learner formulation of Künzel et al. [57]. This pseudo-outcome represents an adjusted version of yield that incorporates (a) the predicted difference between treated and untreated outcomes and (b) a correction term that compensates for any imbalance in treatment assignment. This adjustment ensures that the estimator captures causal effects rather than merely reflecting correlations created by non-random treatment decisions.

Finally, a flexible regression model (such as XGBoost) is trained on these pseudo-outcomes to learn how the treatment effect varies across different values of the covariates. The result is an estimate of the Conditional Average Treatment Effect (CATE) for each plot, reflecting how treatment responsiveness changes with soil properties, plant status, environmental factors, or management practices [52].

Because the pseudo-outcome integrates information from both the outcome and propensity models, the DR-Learner yields a stable, bias-corrected estimate of heterogeneous treatment effects. The doubly robust structure ensures that treatment effect estimates remain consistent as long as either the outcome models or the propensity model has been learned correctly, making the DR-Learner well suited for observational agricultural datasets where confounding is unavoidable and treatment decisions are not randomized.

Because N applications in commercial rice fields were generally high, yield responses showed limited variability across most of the N range, making the modeling of a continuous N–yield relationship challenging. However, yield differences between 240 and 260 kg N ha^{−1} were statistically significant ($p < 0.001$) (Figure 2), indicating meaningful

local heterogeneity around these agronomic decision points. For this reason, a two-anchor doubly robust (DR) learner was implemented for the N_DML model. Two N thresholds (240 and 260 kg N ha⁻¹) were used to define binary treatment indicators, allowing the model to characterize how yield responds when N rates cross these important levels. For each threshold, a propensity model predicting treatment assignment and two outcome models predicting yield under treated and untreated conditions were fitted using the full set of pre-treatment covariates. Doubly robust pseudo-outcomes were then constructed, which are adjusted versions of yield that account for differences in both treatment assignment and expected yield. These corrected values allow the final XGBoost model to learn how N affects yield more accurately. In other words, pseudo-outcomes transform raw data into a form where the ML model can directly learn the causal effect rather than the outcome itself. The two anchor-specific DR models were packaged into a unified bundle and used for subsequent inference and sensitivity analyses. The implementation was based on custom Python code built on XGBoost [58] and scikit-learn [59], following the doubly robust and orthogonal learning frameworks of Chernozhukov et al. [11] and Kennedy [54].

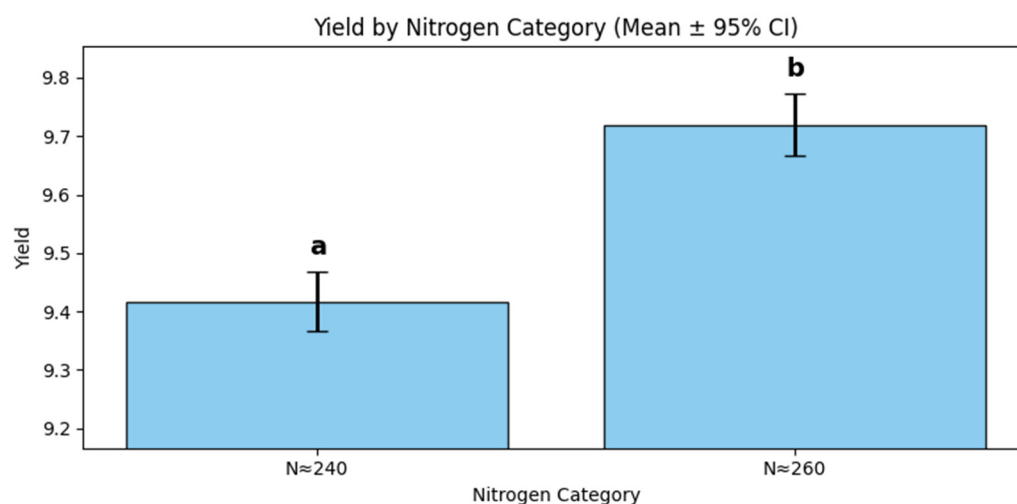


Figure 2. Mean rice yield for two N rate groups centered on two common management levels (N ≈ 240 and N ≈ 260 kg N ha⁻¹), where observations were grouped if N was within ±10 kg ha⁻¹ of each level. Bars represent group means and error bars indicate 95% confidence intervals. Different letters above bars (a, b) denote statistically significant differences between N rate groups based on one-way ANOVA ($p < 0.001$).

2.10.3. Final Feature Space of the DML Models

Because LNC is strongly correlated with N (LNC increases with higher soil N rates), it was removed from the N_DML model [9]. In DML, variables that are caused by the treatment cannot be included as covariates or confounders, as doing so would open post-treatment bias pathways. Therefore, only variables that influence both the treatment and the outcome may be included in the adjustment set. In contrast, for the LNC_DML model, N was not removed from the covariate space, because N is a plausible cause of the LNC rather than a consequence of it. Thus, including N in the adjustment set is appropriate for the LNC_DML model.

The final feature space for the N_DML and the LNC_DML models is referred to Table 1. The selection of covariates (effect modifiers) was guided primarily by agronomic domain knowledge. Variables were included because established crop physiology and field expertise indicate that they influence how N levels affect rice yield.

Table 1. Covariates (X) and confounders (W) which are used in the N_DML and LNC_DML models.

Variable	Covariates (X)	Confounders (W)	Included in Model
Nitrogen	✓	✓	LNC_DML only
Cultivar	✓	✓	N_DML, LNC_DML
year	✓	✓	N_DML, LNC_DML
Radar_Jun_roll	✓	✓	N_DML, LNC_DML
sand	✓	✓	N_DML, LNC_DML
OM	✓	✓	N_DML, LNC_DML
PFP1	✓	✓	N_DML, LNC_DML
soil_resid_N	✓	✓	N_DML, LNC_DML
N_inputs	✓	✓	N_DML, LNC_DML
May_precip	✓	✓	N_DML, LNC_DML
Aug_temp_anom	✓	✓	N_DML, LNC_DML
Jun_temp_anom	✓	✓	N_DML, LNC_DML
May_temp_anom	✓	✓	N_DML, LNC_DML
Days_from_seeding	✓	✓	N_DML, LNC_DML
BCCE *		✓	N_DML, LNC_DML
total_avail_K *		✓	N_DML, LNC_DML
Fe_bind *		✓	N_DML, LNC_DML
Mn_bind *		✓	N_DML, LNC_DML
rec_Mn_dose *		✓	N_DML, LNC_DML
total_avail_P *		✓	N_DML, LNC_DML
Mg_content *		✓	N_DML, LNC_DML
rec_Zn_dose *		✓	N_DML, LNC_DML
B_fix_fac *		✓	N_DML, LNC_DML
K_corr_coeff *		✓	N_DML, LNC_DML
rec_Cu_dose *		✓	N_DML, LNC_DML
Zn_bind *		✓	N_DML, LNC_DML
Mg_corr_coeff *		✓	N_DML, LNC_DML
P_corr_coeff *		✓	N_DML, LNC_DML
Broad_N **		✓	N_DML, LNC_DML

* These variables represent soil-derived indices and correction factors calculated using established agronomic formulations described in detail by Koukoulakis and Papadopoulos [25]. Their inclusion in the final feature set was supported by feature-importance analysis using a Random Forest model for yield prediction, where they were retained as informative predictors. These variables are used exclusively as confounders in the causal models and their full mathematical derivations are provided in the original reference, while a descriptive summary of the variables is included in Appendix A. ** Broad_N refers to broadcast N applied before seeding. Its rate changed slightly during the later years in which the experiment was utilized (an increase of 20 kg ha⁻¹ compared to the first two years); therefore, it was treated as a confounder rather than a covariate in the DML models. ✓ indicates that the variable was included in the corresponding set (X or W).

Cultivar was included because different cultivars are known to have different N requirements. The year variable was added because N needs vary annually depending on weather conditions. For example, high temperatures during flowering can cause yield losses through spikelet sterility [60], but these losses may be partially compensated by higher N application rates due to increased tillering. For the same reason, ECMWF temperature predictions for August, June, and May were included. Hindcast precipitation for May was also included, as Iatrou et al. [9] showed that it is highly important for rice yield. Higher rainfall in May increases water depth in rice fields, reducing N losses through leaching and volatilization. Soil-related covariates such as organic matter (OM), soil residual N (soil_resid_N), and N inputs from mineralization (N_inputs) were included because they represent additional N sources that interact with applied fertilizer. Sand content was also considered, as high sand levels increase water drainage in rice paddies, leading to higher N losses via N₂O and N₂ emissions. Finally, PFP1 [48] was included because the DML models demonstrated improved performance when it was part of the covariate set. PFP1 correlates strongly with clay and sand content and thus captures soil texture-related variation in the

N–yield relationship. The number of days from seeding was included in the covariates because plant developmental stage is strongly related to N requirements.

2.10.4. Model Evaluation and Sensitivity Analysis

Evaluating causal models is challenging because, unlike predictive models, they do not have direct accuracy metrics such as R^2 or Mean Absolute Error (MAE). In contrast, ML prediction tasks have achieved widespread success partly because validation is straightforward [51]. To address this limitation, Facure [12] proposed a method for assessing the sensitivity of causal and predictive models using the following numerical approximation:

$$\delta y(t)/\delta t \approx \frac{y(t+h) - y(t)}{h}$$

where t represents the treatment variable. In our case, t corresponds to N for the N_DML model (with h defined as an additional amount of 20 kg N ha^{−1}), and to LNC for the LNC_DML model (with h defined as a one-class increase in LNC). This method separates field plots into how sensitive they are to N applications. In the current study, the above-mentioned sensitivity equation should be able to figure out in which plots N applications should increase yield.

For evaluation and visualization purposes, plots were ordered from most sensitive to least sensitive, as suggested by Facure [12]. This ordering enables the construction of a Cumulative Sensitivity Curve, which summarizes how yield responsiveness accumulates across the field plots. Mathematically, the cumulative sensitivity at position k in the ranked list can be expressed as:

$$y'(t)_k = \frac{\sum_{i=1}^k (t_i - \hat{t})(y_i - \hat{y})}{\sum_{i=1}^k (t_i - \hat{t})^2}$$

where t_i is the observed N rate for plot i , $\hat{t} = t_i + h$ represents the increased N level used to test sensitivity (with h being the N increment or the LNC class), y_i is the observed yield, and $\hat{y}_i$ is the model-predicted yield at the increased N level or the increased LNC class. The curve summarizes how yield responsiveness accumulates across plots, identifying the areas where additional N is most likely to generate agronomic benefit.

In the CML framework, the quantity of interest is the Conditional Average Treatment Effect (CATE) [52], defined as the expected change in the outcome resulting from a marginal change in the treatment, conditional on observed covariates. These covariates include both confounding variables and effect modifiers that influence the magnitude and direction of treatment effects. Formally, for a given set of covariates $X = x$, the CATE is defined as

$$\text{CATE}(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x],$$

where $Y(1)$ and $Y(0)$ denote the potential outcomes under treatment and control, respectively. In this study, CATE represents the plot-specific yield response to changes in N rate or LNC, considering the soil, climatic, and management characteristics.

2.10.5. Proposed Decision Support System (DSS) Framework

In addition to estimating treatment effects separately, the two models were jointly used to support a practical N recommendation system. The DR-Learner captures local yield gains or losses when N rates move across key agronomic thresholds (240 and 260 kg N ha^{−1}), while the Causal Forest quantifies how the leaf N concentration (LNC) modifies crop responsiveness to fertilization across the full covariate space. By combining these two signals, i.e., threshold-based yield sensitivity from the DR-Learner and plant-status feedback from the Causal Forest, a consistent, rule-based decision framework was created to adjust N

recommendations upward, downward, or maintain them, depending on whether local DR-based yield effects and LNC-based responsiveness were positive, neutral, or negative. This hybrid approach forms the basis of the proposed decision support tool, integrating both agronomic thresholds and continuous plant-status information.

2.10.6. XGBoost Predictive Models

Two predictive (non-causal) models were also fitted: one using the $W + X + T$ feature space of the N_DML model and one using the $W + X + T$ feature space of the LNC_DML model, both with yield (Y) as the target variable. The two rice-yield predictive models, which served as baselines for evaluating the DML models, were based on the XGBoost algorithm. Because the performance of XGBoost is highly sensitive to the choice of hyperparameter, owing to its gradient boosting structure and its potential susceptibility to overfitting, the Optuna library was employed for automated hyperparameter optimization [61]. Since the two predictive models differed only in how LNC was represented (one continuous and one ordinal), Optuna hyperparameter tuning was performed once, and the resulting optimal hyperparameters were applied to both models. The final XGBoost configuration was as follows: $n_estimators = 1902$, $learning_rate = 0.04$, $max_depth = 9$, $subsample = 0.775$, $colsample_bytree = 0.741$, $gamma = 0.726$, $reg_alpha = 0.06$, $reg_lambda = 1.984$, $min_child_weight = 7$, $tree_method = "hist"$, and $max_bin = 185$. SHAP analysis was used to construct feature-importance visualizations for both the XGBoost and DML models. All data processing, model development, and visualizations were implemented in Python using the libraries SHAP (v.0.5), Matplotlib (v.3.10.8), and Seaborn (v.0.13.2) [62–65]. During implementation, a generative AI tool (ChatGPT, OpenAI, GPT-5.1) was used to assist with drafting and debugging portions of the Python code for the DR-Learner framework; all methodological choices, model configurations, and analyses were designed, verified, and validated by the authors.

2.11. Statistical Analysis

Differences in rice yield between N rate groups were evaluated using one-way analysis of variance (ANOVA). Normality was assessed through visual inspection of yield distributions and Q–Q plots. Given the large sample size, minor deviations from normality were considered acceptable, and ANOVA was applied due to its robustness under such conditions.

3. Results

3.1. Nitrogen Effect on Rice Yield

To evaluate whether yield differed between N applications centered around two common management levels, we grouped all observations with N rates within $\pm 10 \text{ kg ha}^{-1}$ of 240 and 260 into two categories ($N \approx 240$ and $N \approx 260$). These categories were then compared using one-way ANOVA. The analysis showed a statistically significant difference in yield between the two N ranges ($p < 0.001$) (Figure 1).

3.2. Performance of the DML Models Compared to the Predictive Models

Figure 3 shows that the DR-Learner outperforms the XGBoost predictive model in identifying yield sensitivity to N. When plots are sorted from most to least responsive to additional N, the DR-Learner curve starts at a higher sensitivity level compared to the predictive model and then gradually declines as it reaches areas where additional N fertilizer is unlikely to increase yield. The early flattening of the curve indicates that, under commercially high N rates, many plots are already sufficiently fertilized and thus show limited potential benefit from additional N applications. Consequently, the DR-Learner identified high-response areas more effectively, showing a strong early rise followed by a

decline as plots with limited yield gain potential were reached. The predictive model shows weaker discrimination, while the random model remains near the average sensitivity.

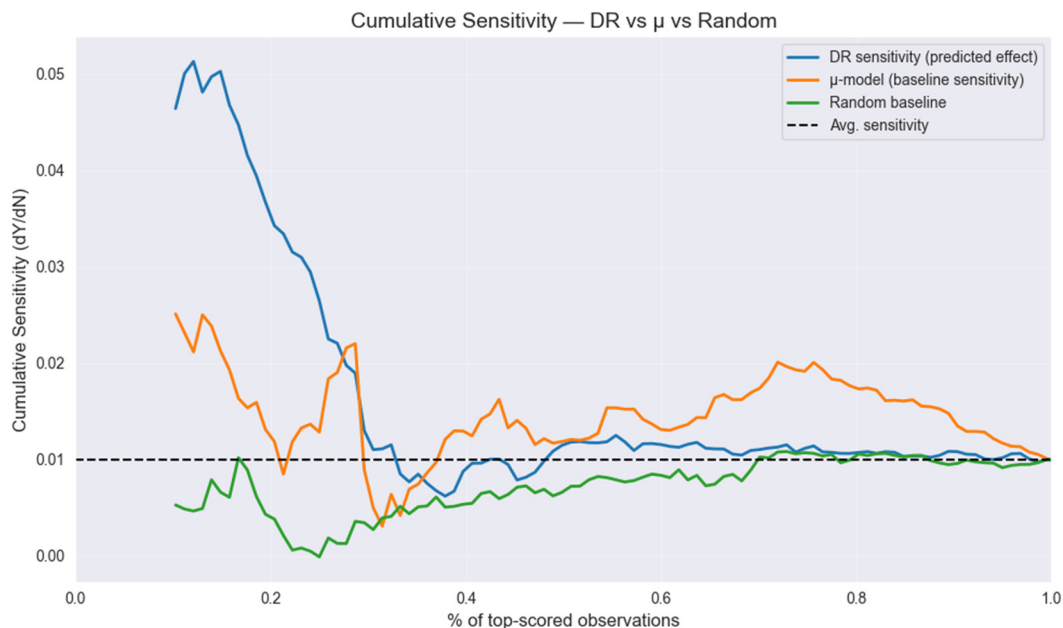


Figure 3. Cumulative sensitivity curves for N responsiveness comparing the causal DR-Learner (N_DML), the predictive XGBoost model (μ -model), and a random baseline. Sensitivity is computed as the model-predicted yield change for a small N increase ($h = +20 \text{ kg N ha}^{-1}$) at each plot, and plots are ranked from most to least sensitive. The cumulative curve shows how responsiveness accumulates as increasingly less-responsive plots are included. Early steep increases indicate better identification of high-response plots.

Average sensitivity was computed by estimating, for each plot with N rate t_i , the yield response to a small N increase h (20 kg N ha^{-1} in this study),

$$\Delta y_i = y(t_i + h) - y(t_i), \quad s_i = \frac{\Delta y_i}{h},$$

and then averaging these individual sensitivities across all plots:

$$\text{Average Sensitivity} = \frac{1}{n} \sum_{i=1}^n s_i. \quad (1)$$

Figure 4 shows that the Causal Forest outperforms the XGBoost predictive model in identifying plots that respond to increases in LNC. It is important to note that the absolute sensitivity values of the N_DML and LNC_DML models are not directly comparable because the treatment increments differ (a 20 kg N ha^{-1} increase for N_DML versus a one-class LNC increase for LNC_DML). The apparent difference in scale between the curves therefore reflects differences in treatment definition rather than model performance. In commercial rice production, N application rates vary only modestly across fields, as growers typically apply uniformly high N doses to compensate for potential N losses. As a result, the marginal yield response to a 20 kg N ha^{-1} increment is small, corresponding to approximately $0.01\text{--}0.05 \text{ t ha}^{-1}$ (i.e., $10\text{--}50 \text{ kg}$ of additional yield). Despite these small marginal effects, the N_DML model still performs well, consistently identifying the plots where this additional yield can be realized. In contrast, LNC varies substantially across fields due to soil heterogeneity, water movement, and differential N losses, which makes it a much more sensitive indicator of plant N status. Consequently, the LNC_DML model captures larger marginal effects and produces cumulative sensitivities in the $0.3\text{--}0.6$ range. It is important to emphasize though that, unlike the

N_DML sensitivity values, which correspond to a change in yield per 20 kg N ha⁻¹ increment, the LNC_DML sensitivities do not correspond to a physical unit of N. A one-class increase in LNC reflects an ordinal shift in plant N status rather than a quantitative fertilizer increment. Therefore, LNC_DML sensitivities cannot be interpreted in kg yield per kg N and should only be compared qualitatively in terms of model ranking ability.

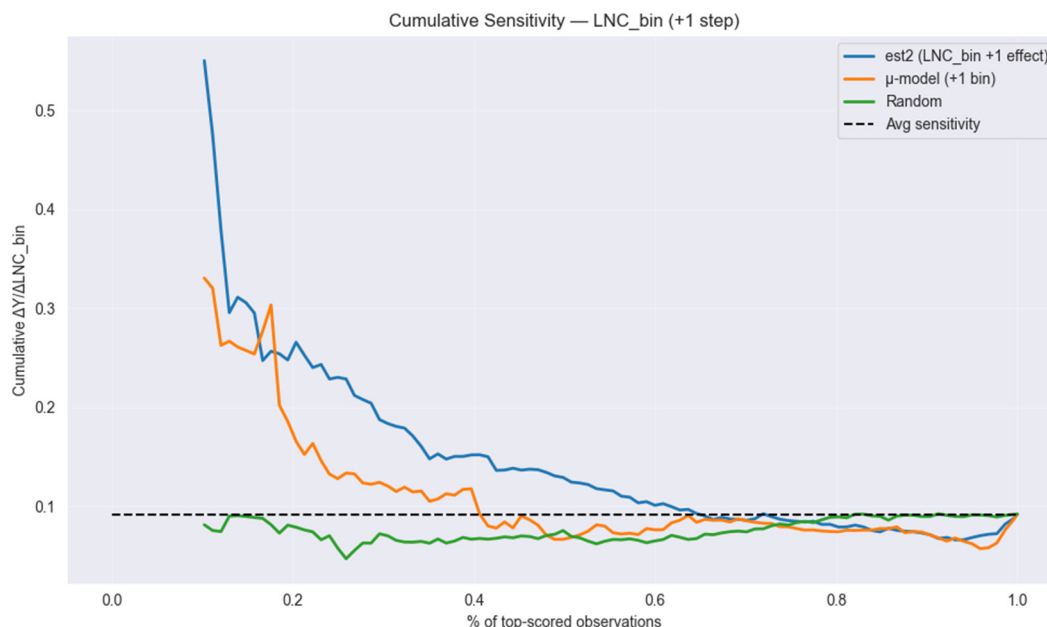


Figure 4. Cumulative sensitivity curves for responsiveness to higher LNC, comparing the causal forest DML model (LNC_DML), the predictive XGBoost model (μ -model), and a random baseline. LNC is discretized into 10 quantile-based classes, and sensitivity is computed as the model-predicted yield change for a one-class increase in LNC. Plots are ranked from most to least sensitive, and cumulative sensitivity summarizes how responsiveness accumulates across ranked plots.

We can also construct SHAP plots for DML models to interpret which variables drive the heterogeneity in the yield response to N rates or increases in LNC. Figure 5 shows that the most influential variables in shaping the N effect were sand content, cultivar, soil organic matter, N inputs from soil supply, residual soil N, days from seeding, and PFP1. These soil- and growth-related factors were followed in importance by the climatic variables, which is expected for rice growing where soil properties and plant development largely determine N responsiveness.

Figure 6 shows the heterogeneity in the yield response to LNC. Overall, the pattern indicates that soil fertility, plant development, and field-specific N dynamics are the primary drivers of heterogeneity in the LNC–yield relationship.

3.3. Decision Support System

Based on the outputs of the N_DML and LNC_DML models, a decision support system (DSS) was developed to assist growers in making informed N management decisions (Table 2). In this framework, the N_DML model provides a baseline recommendation by estimating the CATE of N at agronomically relevant thresholds, indicating whether yield is expected to increase or decrease with changes in N rate. The LNC_DML model complements this information by providing a crop phenotyping signal, estimating the yield response to a one-class increase in discretized LNC, which reflects the physiological N status of the crop.

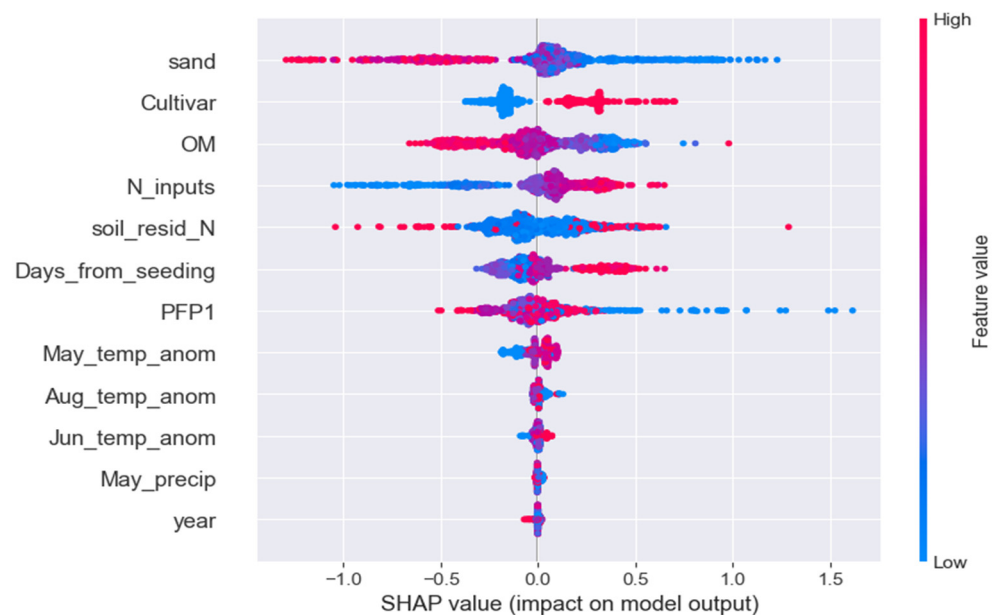


Figure 5. SHAP summary plot showing how each feature contributes to the estimated effect of N on yield in the DR-Learner. Each point represents a plot-level observation, with color indicating the feature value (blue = low; red = high). Features are ordered by their overall importance in explaining heterogeneity in the N effect. Positive SHAP values indicate that the feature increases the predicted yield response to additional N, while negative values indicate a reduced response.

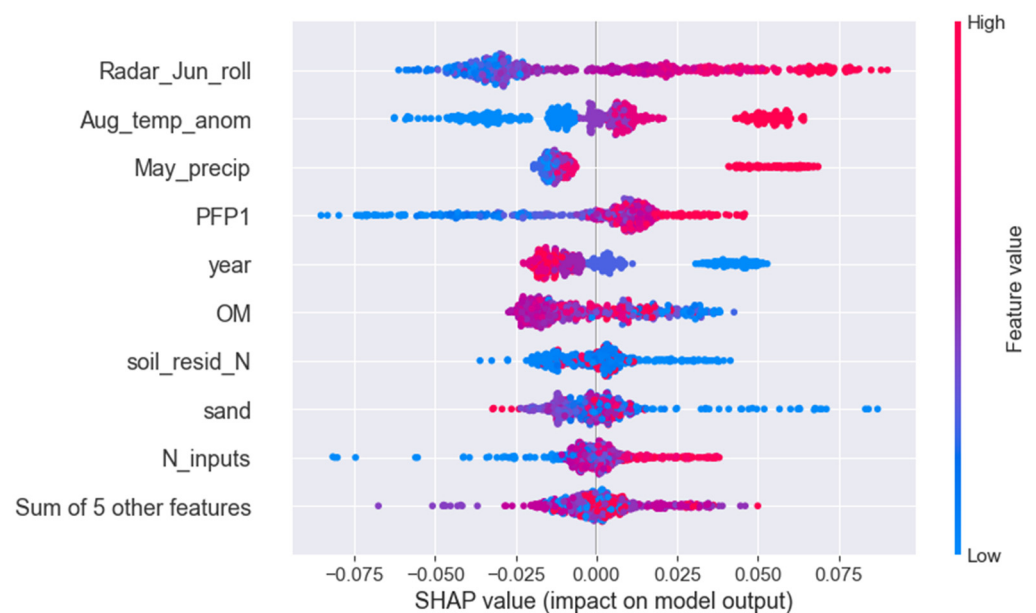


Figure 6. SHAP summary plot showing how each feature contributes to the estimated effect of LNC on yield in the Causal Forest model. Each point represents a plot-level observation, with color indicating the feature value (blue = low; red = high). Features are ordered by their overall importance in explaining heterogeneity in the LNC effect. Positive SHAP values indicate that the feature increases the predicted yield response to higher LNC, whereas negative values indicate a reduced response.

For each grid cell, a positive CATE indicates that increasing N is expected to increase yield, while a negative CATE indicates that reducing N is expected to improve or maintain yield. A CATE equal to zero indicates no expected yield response, and the recommendation is therefore to maintain the current N rate. Final N recommendations are derived by combining the N_DML and LNC_DML signals according to the rules summarized in Table 2, resulting in three agronomically interpretable actions: decrease, maintain, or

increase the N rate. This procedure is applied at the plot level, enabling spatially explicit and biologically informed N management decisions.

Table 2. Combining N_DML and LNC_DML in a DSS.

N_DML (kg N ha ⁻¹)	LNC_DML	Final N Rate (kg ha ⁻¹)
240	positive	260
240	neutral	240
240	negative	220
260	positive	280
260	neutral	260
260	negative	240
280	positive	280
280	neutral	280
280	negative	260

3.4. Predictive Modeling

The XGBoost models trained using the W + X + T feature spaces of the N_DML and LNC_DML frameworks achieved a Mean Absolute Error (MAE) of 0.6 t ha⁻¹ in both cases. Figure 7 presents the corresponding SHAP feature-importance plots. Interestingly, neither organic matter (OM) nor sand appears among the most influential predictors in these purely predictive models. Instead, in the model associated with the N_DML feature space, a climatic variable emerges as the top contributor, suggesting that the predictive model relies more heavily on weather-driven variation than on soil characteristics when estimating yield.

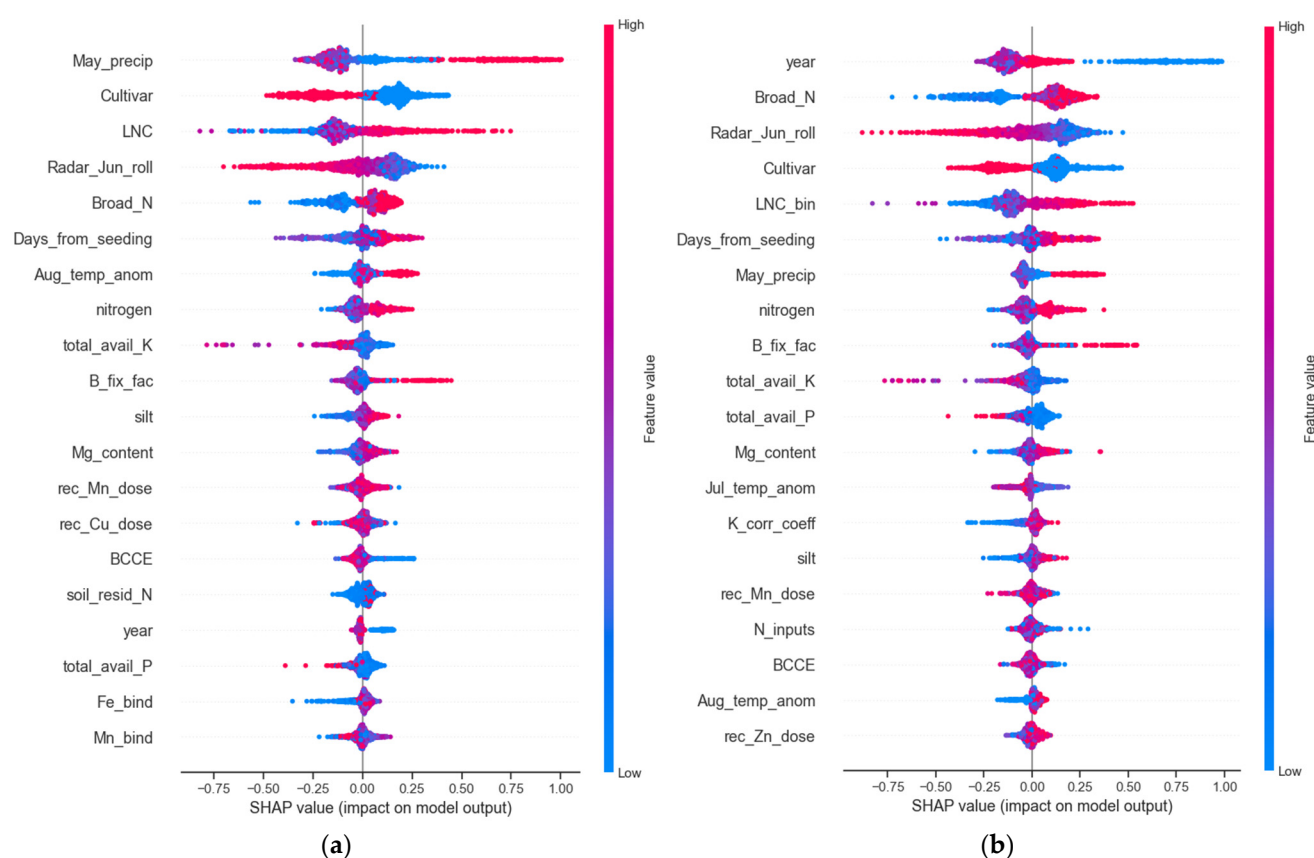


Figure 7. The results of feature evaluation using SHAP for feature importance of the XGBoost model using W + X + T feature space of (a) N_DML and (b) LNC_DML.

4. Discussion

Isolating treatment effects in observational agricultural data collected from commercial fields (i.e., not Randomized Controlled Trials) is inherently challenging because management practices, soil properties, and environmental factors interact strongly [3,66]. Nevertheless, the results of this study show that DML can effectively adjust for confounding and recover the causal effect of N on rice yield under real farm conditions, which is consistent with the DML theory [11,54]. This demonstrates the potential of causal ML to support data-driven fertilization decisions that maximize yield rather than relying on purely correlational patterns, as is often the case with classical predictive ML models.

A substantial limitation of many existing precision-agriculture services is their reliance on simplistic interpretations of vegetation indices such as Normalized Difference Vegetation Index (NDVI), which is known to saturate at moderate-to-high biomass levels and can obscure yield-limiting factors other than N availability [67,68]. For example, fields with low NDVI are often recommended high N doses, while fields with high NDVI are supplied with less N. However, Iatrou et al. [9] showed that this approach can be misleading, because low early-season N uptake may indicate limited yield potential, with very high N inputs at topdressing being detrimental under such conditions. According to Zhao et al. [69], this effect may be attributed to poor grain filling. Such inconsistencies explain why growers frequently mistrust remote-sensing-based recommendations.

Previous advances such as the Variational Autoencoder + LightGBM model by Iatrou et al. [9] achieved a prediction error of 0.6 t ha^{-1} and delivered an average yield increase of 4.32% when deployed in the field. Although promising, this level of improvement is often insufficient for growers to justify the operational effort of variable-rate fertilization, which requires loading shapefiles into the tractor's GPS guidance system, operating multiple fertilizer products (N, P, K), and coordinating several application passes. In practice, the perceived economic gains from reduced fertilizer use are too small to motivate widespread adoption (as established from personal communication with local rice growers).

Agronomically, however, growers in the Axios Plain are technologically advanced and highly equipped, making them well positioned to implement precision fertilization if provided with a system that clearly identifies opportunities for yield improvement, not merely cost savings. The DML-based approach presented in this study addresses this gap. By identifying the specific field plots that are likely to benefit from higher N inputs, without risking overfertilization and yield loss, the model aligns directly with growers' primary objective: increasing yield where possible while avoiding unnecessary N applications elsewhere. This dual benefit has both environmental implications, through reduced N losses in over-supplied areas, and financial implications, by increasing yield in responsive zones and decreasing wasteful spending in non-responsive ones. As such, the model has the potential to substantially improve the practicality and acceptance of precision N management among rice growers.

Applied N rates showed only modest variation across the study fields, as the grower conducting the commercial trials consistently applied relatively high and uniform fertilization rates to offset potential N losses. As a result, the marginal yield response to an additional 20 kg N ha^{-1} was necessarily limited, corresponding to approximately $0.01\text{--}0.05 \text{ t ha}^{-1}$ ($10\text{--}50 \text{ kg}$ of grain). These small values reflect a biological plateau effect rather than a weakness of the N_DML model, which still succeeded in identifying the plots where these incremental gains could be realized. By contrast, LNC exhibited greater variability across fields due to soil heterogeneity, water dynamics, and differences in N loss pathways, making it a more sensitive indicator of crop N status. The LNC_DML model therefore produced larger cumulative sensitivities. However, these values cannot be interpreted in fertilizer units, because a one-class increase in LNC represents an ordinal

shift in plant N status rather than a quantitative increment of applied N. Accordingly, the N_DML and LNC_DML sensitivities should not be compared numerically but rather in terms of their ability to rank plots by responsiveness. Overall, these findings highlight that incorporating physiological indicators such as LNC into causal ML frameworks can greatly enhance the detection of spatial variability in fertilization responsiveness.

The SHAP analyses provided further insight into the agronomic drivers of treatment heterogeneity. For the N_DML model, soil properties (sand, organic matter), soil-derived N inputs, residual soil N, cultivar, and developmental timing (days from seeding) were the dominant factors influencing the yield response to additional N (Figure 5). These variables have well-established roles in determining the efficiency of N uptake and utilization, and their prominence in the SHAP results reinforces the biological consistency of the DML estimations [70,71]. On the other hand, the SHAP results of the predictive models did not highlight organic matter or sand as important features, even though both variables are well known to influence N variability. Instead, in the XGBoost model trained on the N_DML feature space, May precipitation emerged as the dominant predictor (Figure 7a). While May precipitation is strongly correlated with rice yield [9], it is not tightly linked to field-level N heterogeneity, suggesting that the predictive model relies more on climatic covariation than on soil-driven N dynamics.

From a broader modeling perspective, it is important to position the proposed framework relative to recent advances in deep learning for precision agriculture and remote sensing. Numerous studies have demonstrated the strong predictive capabilities of deep learning models, including one-dimensional convolutional neural networks (1D-CNNs), particularly when applied to raw spectral signatures, dense time-series data, or image-based inputs [72,73]. These models are highly effective for pattern recognition tasks and yield prediction under large, well-curated datasets. However, their primary objective is predictive accuracy, and they are not explicitly designed to address confounding or to estimate causal treatment effects. In contrast, the present study addresses a causal inference problem: estimating heterogeneous N treatment effects under real-world farming conditions to support decision making. For this reason, model selection was guided not only by predictive performance, but also by interpretability, robustness to structured tabular data, and suitability for causal effect estimation. Tree-based ensemble models such as XGBoost are widely recognized as strong baselines for agronomic tabular datasets, often matching or outperforming deep learning approaches when raw imagery is not directly used [74]. Moreover, XGBoost integrates naturally within DML frameworks, where it is used to model both the treatment assignment mechanism (propensity model) and the outcome response, enabling estimation of CATE [11]. Overall, this comparison highlights a fundamental distinction between prediction-oriented and effect-oriented modeling paradigms: while deep learning excels at capturing complex nonlinear patterns, CML provides the appropriate framework for translating multimodal agronomic data into robust, interpretable, and actionable management decisions [10].

Importantly, the hybrid decision support framework developed in this study demonstrates how the causal models can be combined to generate practical fertilization recommendations. The DR-Learner captures yield sensitivity around key agronomic thresholds (240 and 260 kg N ha⁻¹), while the Causal Forest integrates continuous plant-status information from LNC. This combination allows for recommendations that reflect both agronomic thresholds and real-time crop physiological conditions.

Despite the encouraging results, several limitations of the present study should be acknowledged, which also point toward important directions for future research. First, the analysis was conducted at the scale of a single commercial rice farm within the Axios River Plain. Although the dataset spans multiple years and captures substantial within-

field spatial variability, the estimated treatment effects inevitably reflect the specific soil, management, and environmental conditions of this production system. Consequently, while the proposed CML framework is methodologically general, the numerical magnitude of the estimated N effects should not be interpreted as universally transferable to other rice-growing regions without local recalibration.

Finally, while this study focused on N management in rice, the proposed CML framework is not crop-specific. Its application to other crops, nutrients, or management interventions represents a promising avenue for future research. By combining causal inference with remote sensing and agronomic knowledge, such approaches have the potential to advance precision agriculture from yield prediction toward robust, effect-based decision making across a wider range of cropping systems.

5. Conclusions

This study demonstrates that CML provides a robust and practical framework for estimating heterogeneous N yield responses under real commercial farming conditions. By moving beyond classical yield-prediction-oriented ML approaches, the proposed DML framework successfully adjusts for confounding inherent in observational agricultural data and recovers causal relationships between N inputs, plant N status, and yield.

Results show that the CML-based models outperform conventional XGBoost predictive models in identifying field plots that are responsive to additional N, highlighting the importance of causal effect estimation for site-specific fertilization decisions. The integration of a DR-Learner for threshold-based N sensitivity estimation with a Causal Forest model using Leaf Nitrogen Concentration (LNC) enables a physiologically informed decision support system that links yield responsiveness with crop N status.

Importantly, the proposed framework identifies zones where yield gains can be achieved through targeted N increases while avoiding unnecessary N applications in non-responsive areas, thereby supporting both economic and environmental objectives. These findings indicate that CML can meaningfully advance precision N management in rice by capturing nonlinear and spatially heterogeneous treatment responses that are not accessible through purely correlational models.

Author Contributions: Conceptualization, M.I. and C.K.; methodology, M.I.; software, M.I.; validation, M.I. and C.K.; formal analysis, M.I.; investigation, C.K.; resources, S.M.; data curation, M.I.; writing—original draft preparation, M.I.; writing—review and editing, M.I. and C.K.; visualization, M.I. and C.K.; supervision, S.M.; project administration, S.M.; funding acquisition, S.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author as the research is ongoing.

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Conflicts of Interest: Author Spiros Mourelatos is the director of Ecodevelopment S.A. Author Christos Karydas is employed by Ecodevelopment S.A. Christos Karydas contributed to the GIS analysis and map preparation. The company provided GIS software and technical support for map preparation. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

N	Nitrogen
DML	Double Machine Learning
LNC	Leaf Nitrogen Concentration
ML	Machine Learning
CML	Causal Machine Learning
MAE	Mean Absolute Error
CATE	Conditional Average Treatment Effect
N_DML	Double Machine Learning model with N as the treatment
LNC_DML	Double Machine Learning model with LNC as the treatment

Appendix A

This appendix provides a concise description of the soil-derived indices and correction factors marked with an asterisk (*) in Table 1, which were included in the Causal Machine Learning models exclusively as confounders. These variables were computed using established agronomic formulations described in detail by Koukoulakis and Papadopoulos [25] and were used to account for soil nutrient availability and related correction effects. Their inclusion in the final feature space was supported by feature-importance analysis using a Random Forest model for yield prediction.

Table A1. Description of soil-derived indices and correction factors included as confounders in the CML models.

Variable	Description
BCCE	Boron correction coefficient for light soils
total_avail_K	Total available potassium
Fe_bind	Iron percent fixation
Mn_bind	Manganese fixation factor
rec_Mn_dose	Manganese dose
total_avail_P	Total available phosphorus
Mg_content	Residual available magnesium
rec_Zn_dose	Corrected zinc dose
B_fix_fac	Boron fixation factor for medium soils
K_corr_coeff	Potassium correction coefficient
rec_Cu_dose	Copper dose
Zn_bind	Zinc fixation factor
Mg_corr_coeff	Magnesium correction coefficient
P_corr_coeff	Phosphorus correction coefficient

References

- Williams, J.F. *Rice Nutrient Management in California*; UCANR Publications: Richmond, CA, USA, 2010.
- Peng, S.; Buresh, R.J.; Huang, J.; Zhong, X.; Zou, Y.; Yang, J.; Wang, G.; Liu, Y.; Hu, R.; Tang, Q.; et al. Improving Nitrogen Fertilization in Rice by Sitespecific N Management. A Review. *Agron. Sustain. Dev.* **2010**, *30*, 649–656. [\[CrossRef\]](#)
- Iatrou, M.; Karydas, C.; Iatrou, G.; Pitsiorlas, I.; Aschonitis, V.; Raptis, I.; Mpetas, S.; Kravvas, K.; Mourelatos, S. Topdressing Nitrogen Demand Prediction in Rice Crop Using Machine Learning Systems. *Agriculture* **2021**, *11*, 312. [\[CrossRef\]](#)
- Iatrou, M.; Karydas, C.; Iatrou, G.; Zartaloudis, Z.; Kravvas, K.; Mourelatos, S. Optimization of Fertilization Recommendation in Greek Rice Fields Using Precision Agriculture. *Agric. Econ. Rev.* **2018**, *19*, 64–75. [\[CrossRef\]](#)
- Bouman, B.A.M.; Lampayan, R.M.; Tuong, T.P. *Water Management in Irrigated Rice: Coping with Water Scarcity*; IRRI: Los Banos, Phillipines, 2007.
- Iatrou, M.; Tziouvalekas, M.; Tsitouras, A.; Evangelou, E.; Noulas, C.; Vlachostergios, D.; Aschonitis, V.; Arampatzis, G.; Metaxa, I.; Karydas, C.; et al. Analyzing the Impact of Storm ‘Daniel’ and Subsequent Flooding on Thessaly’s Soil Chemistry through Causal Inference. *Agriculture* **2024**, *14*, 549. [\[CrossRef\]](#)

7. European Academies Science Advisory Council. *Extreme Weather Events in Europe Preparing for Climate Change Adaptation: An Update on EASAC's 2013 Study*; EASAC: Austria, Vienna, 2018.
8. Furtak, K.; Wolińska, A. The Impact of Extreme Weather Events as a Consequence of Climate Change on the Soil Moisture and on the Quality of the Soil Environment and Agriculture—A Review. *Catena* **2023**, *231*, 107378. [\[CrossRef\]](#)
9. Iatrou, M.; Karydas, C.; Tseni, X.; Mourelatos, S. Representation Learning with a Variational Autoencoder for Predicting Nitrogen Requirement in Rice. *Remote Sens.* **2022**, *14*, 5978. [\[CrossRef\]](#)
10. Athey, S.; Imbens, G.W. Machine Learning Methods That Economists Should Know About. *Annu. Rev. Econom.* **2019**, *11*, 685–725. [\[CrossRef\]](#)
11. Chernozhukov, V.; Chetverikov, D.; Demirer, M.; Duflo, E.; Hansen, C.; Newey, W.; Robins, J. Double/Debiased Machine Learning for Treatment and Structural Parameters. *Econom. J.* **2018**, *21*, C1–C68. [\[CrossRef\]](#)
12. Facure, M. *Causal Inference in Python: Applying Causal Inference in the Tech Industry*; O'Reilly Media, Incorporated: Sebastopol, CA, USA, 2023; ISBN 9781098140250.
13. Fehr, J.; Piccininni, M.; Kurth, T.; Konigorski, S. Assessing the Transportability of Clinical Prediction Models for Cognitive Impairment Using Causal Models. *BMC Med. Res. Methodol.* **2023**, *23*, 187. [\[CrossRef\]](#)
14. Schölkopf, B.; Locatello, F.; Bauer, S.; Ke, N.R.; Kalchbrenner, N.; Goyal, A.; Bengio, Y. Towards Causal Representation Learning. *arXiv* **2021**, arXiv:2102.11107.
15. Fisher, R.A. *The Design of Experiments*; Oliver & Boyd: Oxford, UK, 1935.
16. Cassman, K.G.; Pingali, P.L. Extrapolating Trends from Long-Term Experiments to Farmers' Fields: The Case of Irrigated Rice Systems in Asia. In *Agricultural Sustainability: Economics, Environmental and Statistical Considerations*; John Wiley & Sons: New York, NY, USA, 1995; pp. 36–84.
17. Lobell, D.B.; Burke, M.B. On the Use of Statistical Models to Predict Crop Yield Responses to Climate Change. *Agric. For. Meteorol.* **2010**, *150*, 1443–1452. [\[CrossRef\]](#)
18. Mulder, R.; Singh, A.B.; Hamilton, A.; Das, P.; Outhred, T.; Morris, G.; Bassett, D.; Baune, B.T.; Berk, M.; Boyce, P.; et al. The Limitations of Using Randomised Controlled Trials as a Basis for Developing Treatment Guidelines. *Evid.—Based Ment. Health* **2018**, *21*, 4. [\[CrossRef\]](#)
19. Kluger, D.M.; Owen, A.B.; Lobell, D.B. Combining Randomized Field Experiments with Observational Satellite Data to Assess the Benefits of Crop Rotations on Yields. *Environ. Res. Lett.* **2022**, *17*, 044066. [\[CrossRef\]](#)
20. Giannarakis, G.; Sitokonstantinou, V.; Lorilla, R.S.; Kontoes, C. Towards Assessing Agricultural Land Suitability with Causal Machine Learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, New Orleans, LA, USA, 18–24 June 2022.
21. Deng, L.; Wan, P.; Ye, F.; Zhao, J. Impact of Agricultural Subsidy on Chemical Fertilizer Use: Empirical Evidence of China's Organic-Substitute-Chemical-Fertilizer Policy Based on Double Machine Learning. *PLoS ONE* **2025**, *20*, e0334751. [\[CrossRef\]](#)
22. Wang, L.; Lyu, J.; Zhang, J. Explicating the Role of Agricultural Socialized Services on Chemical Fertilizer Use Reduction: Evidence from China Using a Double Machine Learning Model. *Agriculture* **2024**, *14*, 2148. [\[CrossRef\]](#)
23. Kakimoto, S.; Mieno, T.; Tanaka, T.S.T.; Bullock, D.S. Causal Forest Approach for Site-Specific Input Management via on-Farm Precision Experimentation. *Comput. Electron. Agric.* **2022**, *199*, 107164. [\[CrossRef\]](#)
24. Litskas, V.D.; Aschonitis, V.G.; Lekakis, E.H.; Antonopoulos, V.Z. Effects of Land Use and Irrigation Practices on Ca, Mg, K, Na Loads in Rice-Based Agricultural Systems. *Agric. Water Manag.* **2014**, *132*, 30–36. [\[CrossRef\]](#)
25. Koukoulakis, P.; Papadopoulos, A. *Η Ερμηνεία Της Ανάλυσης Του Εδάφους*; Stamoulis: Athens, Greece, 2001.
26. Iatrou, M.; Tziachris, P.; Biliadis, F.; Kekelis, P.; Pavlakis, C.; Theofilidou, A.; Papadopoulos, I.; Strouthopoulos, G.; Giannopoulos, G.; Arampatzis, D.; et al. Data-Driven and Mechanistic Soil Modeling for Precision Fertilization Management in Cotton. *Nitrogen* **2025**, *6*, 29. [\[CrossRef\]](#)
27. Evans, J.R. Photosynthesis and Nitrogen Relationships in Leaves of C₃ Plants. *Oecologia* **1989**, *78*, 9–19.
28. Jones, J.B. *Laboratory Guide for Conducting Soil Tests and Plant Analysis*; Taylor & Francis: Abingdon, UK, 2001.
29. Magdoff, F.R.; Jokela, W.E.; Fox, R.H.; Griffin, G.F. A Soil Test for Nitrogen Availability in the Northeastern United States. *Commun. Soil Sci. Plant Anal.* **1990**, *21*, 1103–1115. [\[CrossRef\]](#)
30. Walkley, A.; Black, I.A. An examination of the Degtjareff method for determining soil organic matter, and a proposed modification of the chromic acid titration method. *Soil Sci.* **1934**, *37*, 29–38.
31. van Reeuwijk, L.P. *Procedures for Soil Analysis*; International Soil and Information Center: Wageningen, The Netherlands, 2002.
32. Bouyoucos, G.J. Hydrometer Method Improved for Making Particle Size Analyses of Soils¹. *Agron. J.* **1962**, *54*, 464–465.
33. Iatrou, M.; Papadopoulos, A.; Papadopoulos, F.; Dichala, O.; Psoma, P.; Bountla, A. Determination of Soil Available Phosphorus Using the Olsen and Mehlich 3 Methods for Greek Soils Having Variable Amounts of Calcium Carbonate. *Commun. Soil Sci. Plant Anal.* **2014**, *45*, 2207–2214. [\[CrossRef\]](#)
34. Knudsen, D.; Peterson, G.A.; Pratt, P.F. Lithium, Sodium, and Potassium. In *Methods of Soil Analysis*; Agronomy Monographs; John Wiley and Sons: Hoboken, NJ, USA, 1983; pp. 225–246. ISBN 9780891189770.

35. Iatrou, M.; Papadopoulos, A.; Papadopoulos, F.; Dichala, O.; Psoma, P.; Bountla, A. Determination of Soil-Available Micronutrients Using the DTPA and Mehlich 3 Methods for Greek Soils Having Variable Amounts of Calcium Carbonate. *Commun. Soil Sci. Plant Anal.* **2015**, *46*, 1905–1912. [\[CrossRef\]](#)
36. Jeffrey, A.J.; McCallum, L.E. Investigation of a Hot 0.01 m CaCl₂ Soil Boron Extraction Procedure Followed by ICP-AES Analysis. *Commun. Soil Sci. Plant Anal.* **1988**, *19*, 663–673. [\[CrossRef\]](#)
37. Du, G.; Liu, W.; Pan, T.; Yang, H.; Wang, Q. Cooling Effect of Paddy on Land Surface Temperature in Cold China Based on MODIS Data: A Case Study in Northern Sanjiang Plain. *Sustainability* **2019**, *11*, 5672. [\[CrossRef\]](#)
38. Skofronick-Jackson, G.; Petersen, W.A.; Berg, W.; Kidd, C.; Stocker, E.F.; Kirschbaum, D.B.; Kakar, R.; Braun, S.A.; Huffman, G.J.; Iguchi, T.; et al. The Global Precipitation Measurement (GPM) Mission for Science and Society. *Bull. Am. Meteorol. Soc.* **2017**, *98*, 1679–1695. [\[CrossRef\]](#)
39. Espino, L.; Leinfelder-Miles, M.; Brim-Deforest, W.; Al-khatib, K.; Linquist, B.; Swett, C.; McKenzie, K. *Rice Production Manual*; University of California, Agriculture and Natural Resources: Davis, CA, USA, 2018.
40. Stroppiana, D.; Fava, F.; Boschetti, M.; Brivio, P. Estimation of Nitrogen Content in Crops and Pastures Using Hyperspectral Vegetation Indices. In *Hyperspectral Remote Sensing of Vegetation*; CRC: Boca Raton, FL, USA, 2011; pp. 245–262. ISBN 978-1-4398-4537-0.
41. Karydas, C. Temporal Dimensions in Rice Crop Spectral Profiles. *J. Geomat.* **2016**, *10*, 140.
42. Karydas, C.; Iatrou, M.; Mourelatos, S. Yield Data Management in Rice Cultivation in Precision Agriculture Terms: The Greek Paradigm. *AgriEngineering* **2025**, *7*, 413. [\[CrossRef\]](#)
43. Fleming, K.L.; Westfall, D.G.; Wiens, D.W.; Brodahl, M.C. Evaluating Farmer Defined Management Zone Maps for Variable Rate Fertilizer Application. *Precis. Agric.* **2000**, *2*, 201–215. [\[CrossRef\]](#)
44. Heijting, S.; de Bruin, S.; Bregt, A.K. The Arable Farmer as the Assessor of Within-Field Soil Variation. *Precis. Agric.* **2011**, *12*, 488–507. [\[CrossRef\]](#)
45. Kumar, L.; Mutanga, O. Google Earth Engine Applications Since Inception: Usage, Trends, and Potential. *Remote Sens.* **2018**, *10*, 1509. [\[CrossRef\]](#)
46. Gorelick, N.; Hancher, M.; Dixon, M.; Ilyushchenko, S.; Thau, D.; Moore, R. Google Earth Engine: Planetary-Scale Geospatial Analysis for Everyone. *Remote Sens. Environ.* **2017**, *202*, 18–27. [\[CrossRef\]](#)
47. Amani, M.; Ghorbanian, A.; Ahmadi, S.A.; Kakooei, M.; Moghimi, A.; Mirmazloumi, S.M.; Moghaddam, S.H.A.; Mahdavi, S.; Ghahremanloo, M.; Parsian, S.; et al. Google Earth Engine Cloud Computing Platform for Remote Sensing Big Data Applications: A Comprehensive Review. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2020**, *13*, 5326–5350. [\[CrossRef\]](#)
48. Iatrou, M.; Papadopoulos, A. Machine Learning vs. Langmuir: A Multioutput XGBoost Regressor Better Captures Soil Phosphorus Adsorption Dynamics. *Crops* **2025**, *5*, 55. [\[CrossRef\]](#)
49. Breiman, L. Random Forests. *Mach. Learn.* **2001**, *45*, 5–32. [\[CrossRef\]](#)
50. Howard, J.; Gugger, S. *Deep Learning for Coders with Fastai and PyTorch*; O'Reilly Media: Sebastopol, CA, USA, 2020.
51. Hastie, T.; Tibshirani, R.; Friedman, J. *The Elements of Statistical Learning*; Springer New York Inc.: New York, NY, USA, 2001.
52. Lee, S.; Okui, R.; Whang, Y.-J. Doubly Robust Uniform Confidence Band for the Conditional Average Treatment Effect Function. *J. Appl. Econom.* **2017**, *32*, 1207–1225. [\[CrossRef\]](#)
53. Wager, S.; Athey, S. Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests. *J. Am. Stat. Assoc.* **2018**, *113*, 1228–1242. [\[CrossRef\]](#)
54. Kennedy, E.H. Towards Optimal Doubly Robust Estimation of Heterogeneous Causal Effects. *Electron. J. Stat.* **2020**, *17*, 3008–3049.
55. Foster, D.J.; Syrgkanis, V. Orthogonal Statistical Learning. *Ann. Statist.* **2023**, *51*, 879–908.
56. Battocchi, K.; Dillon, E.; Hei, M.; Lewis, G.; Oka, P.; Oprescu, M.; Syrgkanis, V. EconML: A Python Package for ML-Based Heterogeneous Treatment Effects Estimation. 2019. Available online: <https://Github.Com/Py-Why/EconML> (accessed on 22 August 2025).
57. Künzel, S.R.; Sekhon, J.S.; Bickel, P.J.; Yu, B. Metalearners for Estimating Heterogeneous Treatment Effects Using Machine Learning. *Proc. Natl. Acad. Sci. USA* **2019**, *116*, 4156–4165. [\[CrossRef\]](#)
58. Chen, T.; Guestrin, C. XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22Nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, 13–17 August 2016; ACM: New York, NY, USA, 2016; pp. 785–794.
59. Varoquaux, G.; Buitinck, L.; Louppe, G.; Grisel, O.; Pedregosa, F.; Mueller, A. Scikit-Learn. *GetMobile Mob. Comput. Commun.* **2015**, *19*, 29–33. [\[CrossRef\]](#)
60. Shi, P.; Zhu, Y.; Tang, L.; Chen, J.; Sun, T.; Cao, W.; Tian, Y. Differential Effects of Temperature and Duration of Heat Stress during Anthesis and Grain Filling Stages in Rice. *Environ. Exp. Bot.* **2016**, *132*, 28–41. [\[CrossRef\]](#)
61. Akiba, T.; Sano, S.; Yanase, T.; Ohta, T.; Koyama, M. Optuna: A Next-Generation Hyperparameter Optimization Framework. *arXiv* **2019**. [\[CrossRef\]](#)
62. Joseph, A. *Shapley Regressions: A Framework for Statistical Inference on Machine Learning Models*, 4th ed.; King's College London: London, UK, 2019.

63. Hunter, J.D. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng.* **2007**, *9*, 90–95. [[CrossRef](#)]
64. Waskom, M.; Botvinnik, O.; O’Kane, D.; Hobson, P.; Lukauskas, S.; Gemperline, D.C.; Augspurger, T.; Halchenko, Y.; Cole, J.B.; Warmenhoven, J.; et al. *Mwaskom/Seaborn: V0.8.1 (September 2017)*; Zenodo: Geneva, Switzerland, 2017. [[CrossRef](#)]
65. Van Rossum, G.; Drake, F.L. Python Tutorial. *History* **2010**, *42*, 1–122. [[CrossRef](#)]
66. Cassman, K.; Dobermann, A.; Walters, D. Agroecosystems, Nitrogen-Use Efficiency, and Nitrogen Management. *Ambio* **2002**, *31*, 132–140. [[CrossRef](#)]
67. Peng, Y.; Gitelson, A.A.; Keydan, G.; Rundquist, D.C.; Moses, W. Remote Estimation of Gross Primary Production in Maize and Support for a New Paradigm Based on Total Crop Chlorophyll Content. *Remote Sens. Environ.* **2011**, *115*, 978–989. [[CrossRef](#)]
68. Sibley, A.M.; Grassini, P.; Thomas, N.E.; Cassman, K.G.; Lobell, D.B. Testing Remote Sensing Approaches for Assessing Yield Variability among Maize Fields. *Agron. J.* **2014**, *106*, 24–32. [[CrossRef](#)]
69. Zhao, C.; Liu, G.; Chen, Y.; Jiang, Y.; Shi, Y.; Zhao, L.; Liao, P.; Wang, W.; Xu, K.; Dai, Q.; et al. Excessive Nitrogen Application Leads to Lower Rice Yield and Grain Quality by Inhibiting the Grain Filling of Inferior Grains. *Agriculture* **2022**, *12*, 962. [[CrossRef](#)]
70. Fageria, N.K.; Baligar, V.C. Enhancing Nitrogen Use Efficiency in Crop Plants. In *Advances in Agronomy*; Academic Press: Cambridge, MA, USA, 2005; Volume 88, pp. 97–185, ISBN 0065-2113.
71. Iatrou, M.; Papadopoulos, A. Influence of Nitrogen Nutrition on Yield and Growth of an Everbearing Strawberry Cultivar (Cv. Evie II). *J. Plant Nutr.* **2016**, *39*, 1499–1505. [[CrossRef](#)]
72. Kamilaris, A.; Prenafeta-Boldú, F.X. Deep Learning in Agriculture: A Survey. *Comput. Electron. Agric.* **2018**, *147*, 70–90. [[CrossRef](#)]
73. Zhang, L.; Zhang, L.; Du, B. Deep Learning for Remote Sensing Data: A Technical Tutorial on the State of the Art. *IEEE Geosci. Remote Sens. Mag.* **2016**, *4*, 22–40. [[CrossRef](#)]
74. Grinsztajn, L.; Oyallon, E.; Varoquaux, G. Why Do Tree-Based Models Still Outperform Deep Learning on Tabular Data? *Adv. Neural Inf. Process. Syst.* **2022**, *35*, 507–520. [[CrossRef](#)]

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